PORTER

Delivery Time Prediction

LINEAR REGRESSION MODEL

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DS C77

Delivery Time Prediction using Linear Regression

Problem Statement

Objective

This assignment will help you apply linear regression techniques to a real-world business scenario. By working through this exercise, you will gain hands-on experience in using regression analysis to identify key relationships between variables, make data-driven predictions and extract actionable business insights. You will also develop an understanding of how to interpret regression outputs, assess model performance and effectively communicate findings to support strategic decision-making.

Business Value

The growing demand for quick and efficient delivery in the logistics industry calls for the development of systems that can predict delivery times accurately. Porter, an intra-city logistics marketplace, services millions of customers daily, and optimising delivery times is crucial for improving operational efficiency. The objective is to build a linear regression model that can perform the following:1. Predict the delivery time for an order based on various input features.2. Help optimise delivery operations by providing accurate time estimates.3. Support operational planning and resource management, allowing for more effective use of delivery partners

Dataset Overview

Context

The Porter Parcel Delivery Times data set provides details about orders placed through Porter. The data has information about the orders, such as item quantity and order amount; information about the store, such as category, distance from customer and market ID; and operational information, such as the number of delivery partners.

Content

The dataset consists of multiple observations, each representing data for a specific order, with the following key attributes:

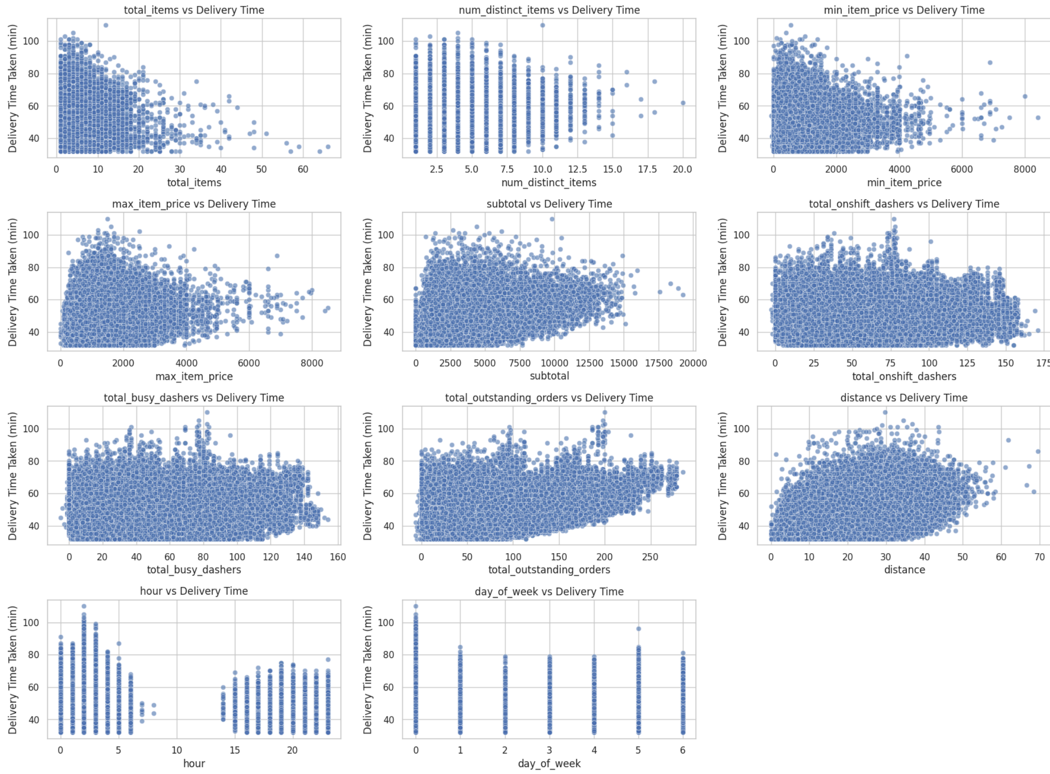
Column

Column Name	Description
market_id	ID of the market where the restaurant is located.
created_at	The timestamp when the order was placed.
actual_delivery_time	The timestamp when the order was delivered.
store_primary_category	The category of the restaurant (e.g., fast food or dine-in).
order_protocol	The integer code indicating how the order was placed (e.g., via Porter or call to restaurant).
total_items	The total number of items in the order.
Subtotal	The total price of the order
num_distinct_items	The number of distinct items in the order
min_item_price	The price of the cheapest item in the order
max_item_price	The price of the most expensive item in the order
total_onshift_dashers	The number of delivery partners on duty at the time the order was placed
total_busy_dashers	The number of delivery partners already attending to other tasks
total_outstanding_orders	The number of outstanding orders to be fulfilled at the time the order was placed
distance	The total distance from the restaurant to the customer

Visual Explorations and Insights

Below are visualizations that helped understand distributions, relationships, and patterns in the data, along with interpretations.

Scatter plots of numerical features vs delivery time.

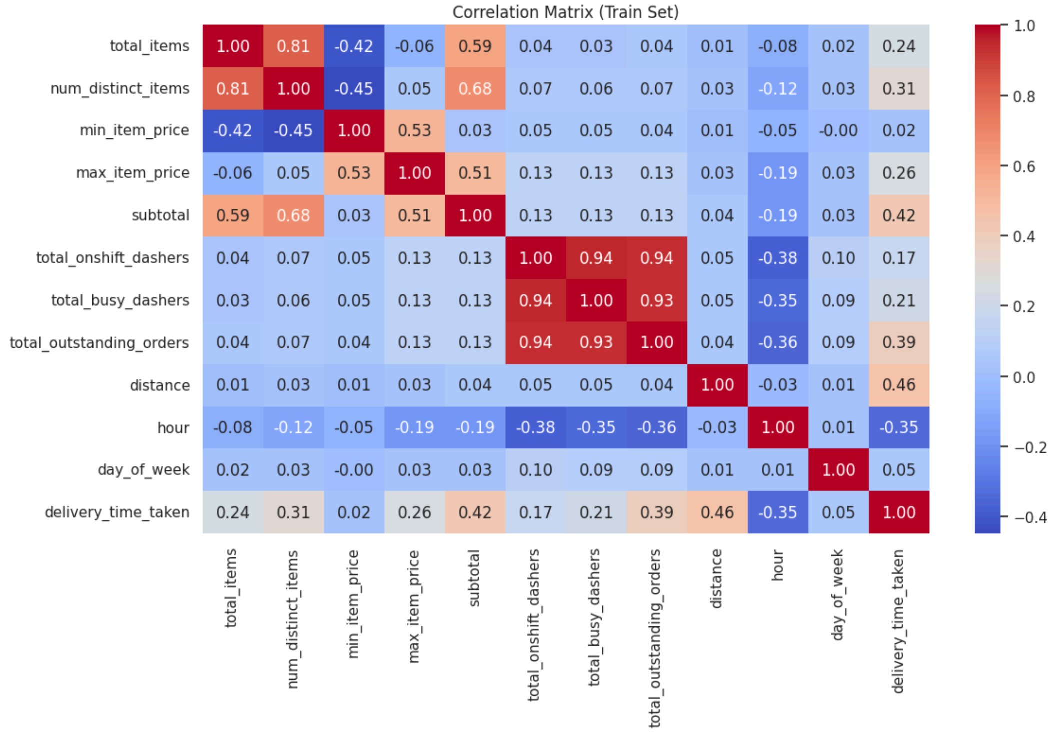


Scatter plots of numerical features vs delivery time.

Interpretation:

Distance , outstanding orders and busy dashers contribute to an increase in delivery time,
Also there is an increase in delivery time during early hours.
More on shift dashers and higher minimum item price leads to a decrease in delivery time.

Correlation heatmap to identify strongest predictors.

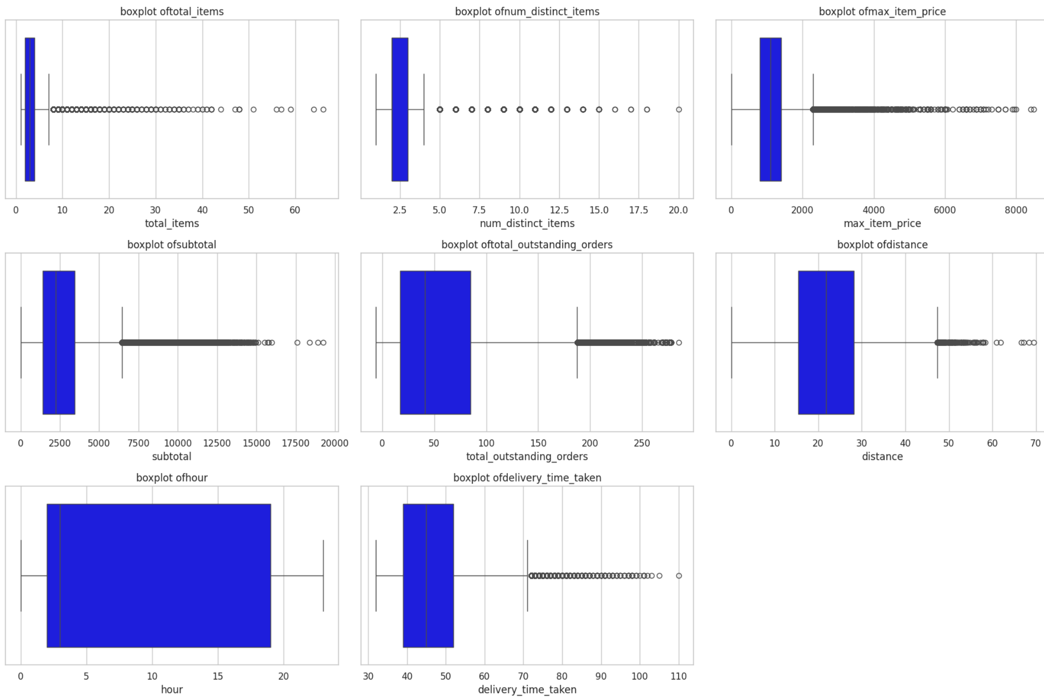


	total_items	num_distinct_items	min_item_price	max_item_price	subtotal	total_onshift_dashers	total_busy_dashers	total_outstanding_orders	distance	hour	day_of_week	delivery_time_taken
total_items	1.00	0.81	-0.42	-0.06	0.59	0.04	0.03	0.04	0.01	-0.08	0.02	0.24
num_distinct_items	0.81	1.00	-0.45	0.05	0.68	0.07	0.06	0.07	0.03	-0.12	0.03	0.31
min_item_price	-0.42	-0.45	1.00	0.53	0.03	0.05	0.05	0.04	0.01	-0.05	-0.00	0.02
max_item_price	-0.06	0.05	0.53	1.00	0.51	0.13	0.13	0.13	0.03	-0.19	0.03	0.26
subtotal	0.59	0.68	0.03	0.51	1.00	0.13	0.13	0.13	0.04	-0.19	0.03	0.42
total_onshift_dashers	0.04	0.07	0.05	0.13	0.13	1.00	0.94	0.94	0.05	-0.38	0.10	0.17
total_busy_dashers	0.03	0.06	0.05	0.13	0.13	0.94	1.00	0.93	0.05	-0.35	0.09	0.21
total_outstanding_orders	0.04	0.07	0.04	0.13	0.13	0.94	0.93	1.00	0.04	-0.36	0.09	0.39
distance	0.01	0.03	0.01	0.03	0.04	0.05	0.05	0.04	1.00	-0.03	0.01	0.46
hour	-0.08	-0.12	-0.05	-0.19	-0.19	-0.38	-0.35	-0.36	-0.03	1.00	0.01	-0.35
day_of_week	0.02	0.03	-0.00	0.03	0.03	0.10	0.09	0.09	0.01	0.01	1.00	0.05
delivery_time_taken	0.24	0.31	0.02	0.26	0.42	0.17	0.21	0.39	0.46	-0.35	0.05	1.00

Interpretation:

Distance , total outstanding orders, subtotal have the highest positive correlation and hour of day has a negative correlation

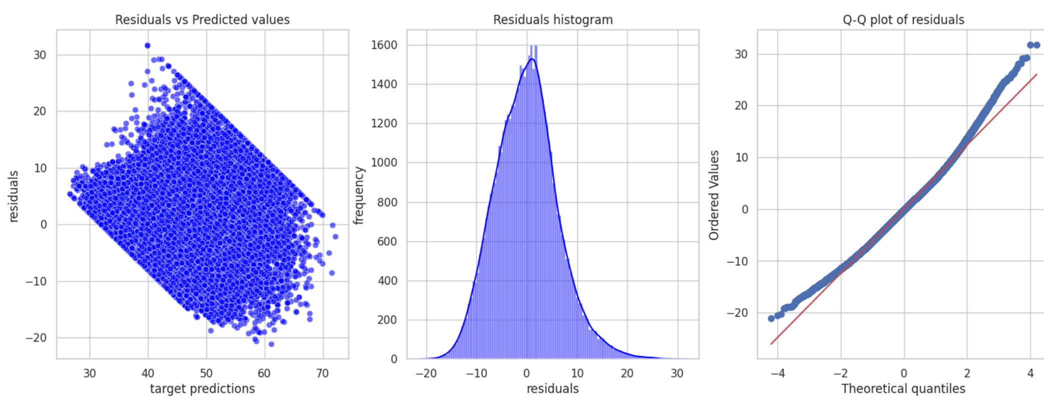
Boxplots showing outliers in delivery time and other numerical features



Interpretation:

Almost all features have outliers except hours which were handled with the Inter quantile range method.

Residual plots showing model fit.



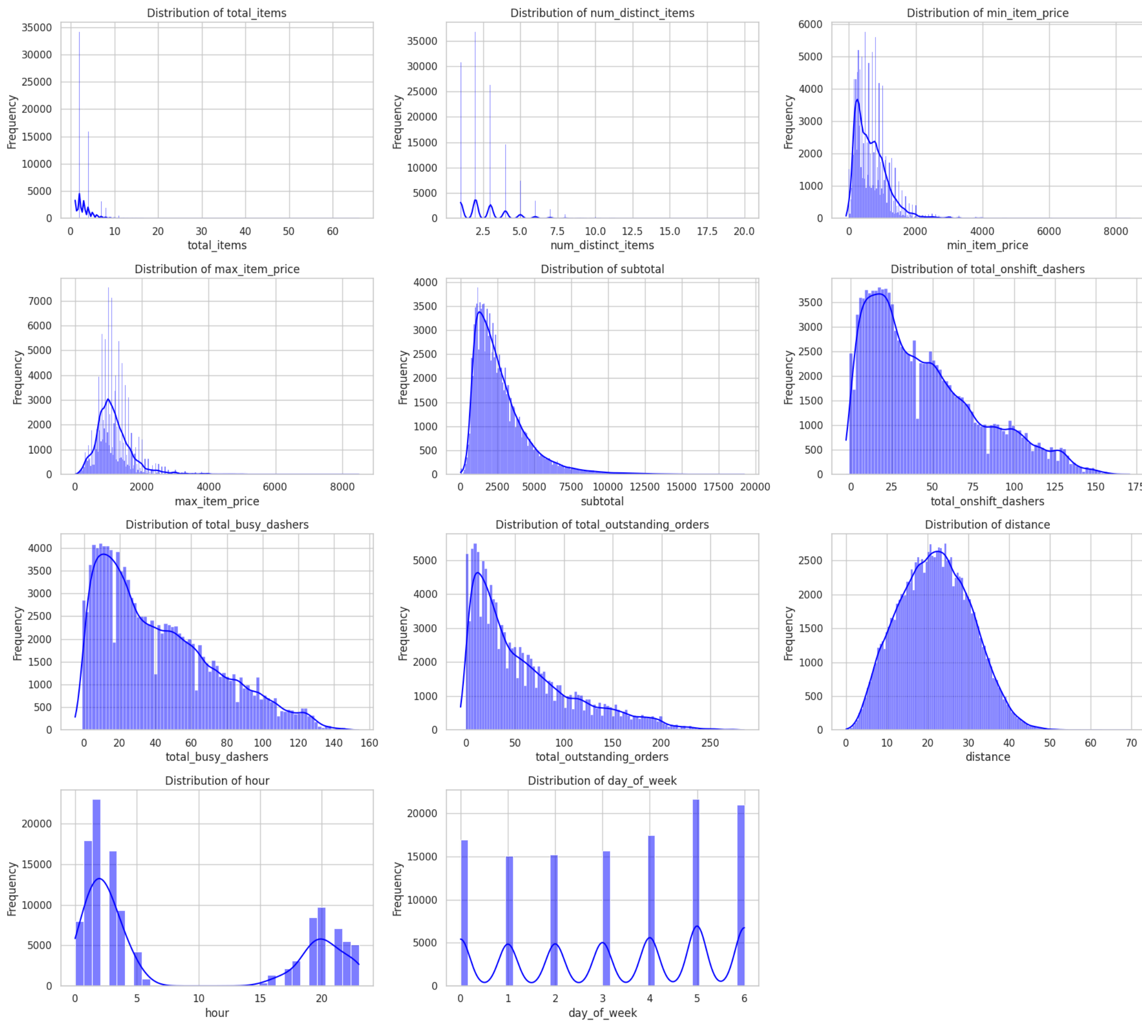
Interpretation:

Residuals vs PredictedValuesshowsfanshapedpatternindicatingheteroscedasticity.Residuals Histograms is roughly bell shaped and centered at zero showing residuals are normally distributed. Q-Q plot shows that points follow the line but deviates at tails

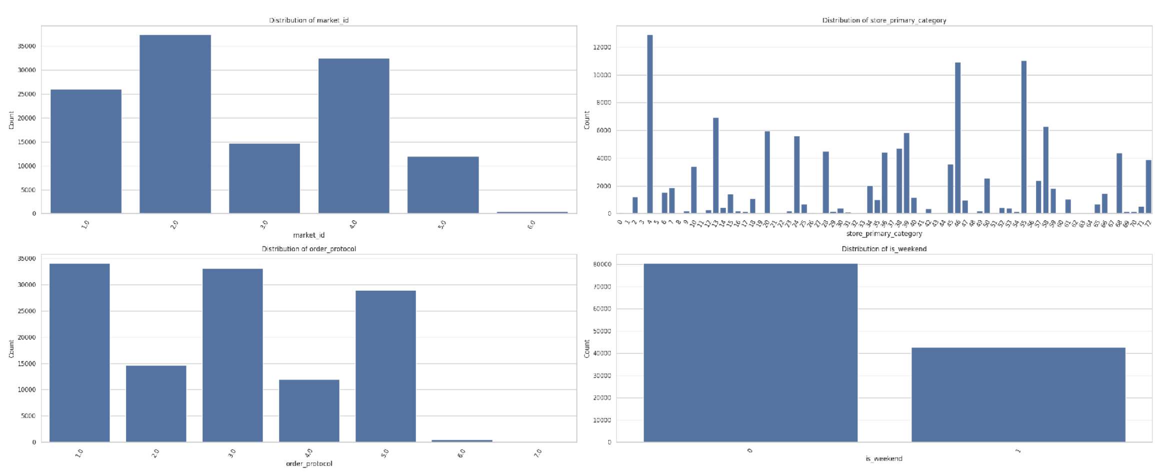
Distribution of delivery times (target variable).

Interpretation: Majority of deliveries are within 35-50 minutes , whereas the peak is between40-45 minutes.

Plot distributions for numerical columns in the training set



Plot distribution of categorical features



Results and Insights

Model performance-

Initial model created using stats models with 11 features

OLS Regression Results

Dep. Variable: delivery_time_taken R-squared: 0.538

Model: OLS Adj. R-squared: 0.538

Method: Least Squares F-statistic: 1.302e+04

Date: Sat, 12 Jul 2025 Prob (F-statistic): 0.00

Time: 10:01:10 Log-Likelihood: -3.9918e+05

No. Observations: 123043 AIC: 7.984e+05

Df Residuals: 123031 BIC: 7.985e+05

Df Model: 11

Covariance Type: nonrobust

	coef	std err	t	P>|t|	[0.025	0.975]
const	50.0828	0.062	806.671	0.000	49.961	50.204
market_id	-0.6971	0.013	-52.217	0.000	-0.723	-0.671
store_primary_category	0.0021	0.001	2.465	0.014	0.000	0.004
order_protocol	-0.9263	0.012	-77.497	0.000	-0.950	-0.903
total_items	0.3063	0.045	6.813	0.000	0.218	0.394
subtotal	2.3281	0.038	61.502	0.000	2.254	2.402
num_distinct_items	0.3388	0.038	8.953	0.000	0.265	0.413
max_item_price	0.1104	0.028	3.896	0.000	0.055	0.166
total_outstanding_orders	2.4731	0.020	126.139	0.000	2.435	2.512
distance	3.9889	0.018	225.067	0.000	3.954	4.024
hour	-1.5467	0.019	-79.397	0.000	-1.585	-1.509
is_weekend	1.7417	0.037	46.448	0.000	1.668	1.815

Omnibus: 2499.241 Durbin-Watson: 1.996

Prob(Omnibus): 0.000 Jarque-Bera (JB): 2811.444

Skew: 0.317 Prob(JB):0.00

Kurtosis: 3.383 Cond. No. 150.

Final model with 8 features made using scikit learn and recursive feature elimination

R^2 Score: 0.5358
The model explains ~53.6% of the variance in delivery time, which is reasonable for real-world delivery data affected by many dynamic factors.

RMSE: 6.21 minutes
On average, the model's predictions deviate by about 6.2 minutes from the actual delivery time.

Top 3 Influential Features (based on coefficients):

1. Distance
   - Positive impact on delivery time
   - Intuition: Longer delivery distances lead to higher delivery times.
2. Total Outstanding Orders
   - Positive impact
   - Intuition: If many orders are pending, it delays new deliveries.
3. Subtotal (order value)
   - Positive correlation
   - Larger orders tend to take more time (e.g., cooking time, packaging).

Observations and suggestions

- Order hour and is_weekend also had moderate influence, suggesting time of day and weekend load affect delivery delays.
- Categorical features like market_id and order_protocol had varying impacts, but were encoded numerically and retained based on RFE selection.
- Include external features like weather, traffic, or to make new features by feature engineering to enhance predictions.
- Could use a non-linear model

Subjective Questions

Question 1. Are there any categorical variables in the data? From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Answer: Yes, there are 4 categorical variables in our dataset:

market_id- In our model it had a negative coefficient ,which could mean certain markets have a shorter delivery time than others.

store_primary_category - This variable had many unique values and because of its high cardinality it was not useful in the model.

order_protocol - In our model this variable had a negative coefficient , which suggests that some order methods may be faster than others.

is_weekend - In our model this variable had a positive and significant coefficient indicating longer delivery times on weekend due to high demand or traffic.

Question 2. What does test_size = 0.2 refer to during splitting the data into training and test sets?

Answer: 'test_size = 0.2' means that we have splitted our data in a way that 80% of the data will be used for training and 20% of the data will be used for testing.

Question 3. Looking at the heatmap, which one has the highest correlation with the target variable?

Answer: Distance has the highest correlation of 0.46 with the target variable.

Question 4. What was your approach to detect the outliers? How did you address them?

Answer: I used boxplots to visualize the numerical features and to detect the outliers.To handle the outliers i capped them based on the IQR method

Question 5. Based on the final model, which are the top 3 features significantly affecting the delivery time?

Answer: Based on the final model the top 3 features are distance , is_weekend and num_distinct_items

Question 6. Explain the linear regression algorithm in detail

Answer: Linear Regression is a supervised learning algorithm used for predicting a continuous numeric value based on one or more input features.It assumes a linear relationship between the independent variables and the dependent variable.The model tries to fit a straight line or a hyperplane that best predicts the target variable.

The formula for linear regression is $Y = a + b1\ X1 + b2\ X2 + \ldots + bnXn$

Linear regression tries to find the best value for b1,b2,bn by minimizing the difference between actual and predicted values for the target variable.this difference is measured by mean square error and the goal is to minimize the ean squared error to find the line that best fits the data

Question 7. Explain the difference between simple linear regression and multiple linear regression

Answer: Simple linear regression involves only one independent variable and one dependent variable and models a straight line relationship between them ,whereas Multiple linear regression involves two or more independent variable to predict the dependent variable and models a hyperplane in the multidimensional space

Question 8. What is the role of the cost function in linear regression, and how is it minimized?

Answer: The cost function in linear regression measures how well the model's predictions match the actual values. In linear regression we use mean squared error as the cost function and it is minimized by using methods such as Gradient descent. The algorithm adjusts the coefficients to reduce the cost step by step.

Question 9. Explain the difference between overfitting and underfitting.

Answer: In the case of Overfitting the model learns the training data too well, including noise and outliers. It performs very well on training data but poorly on testing data.whereas In the case of Underfitting The model is too simple to capture the underlying patterns in the data and performs poorly on both training and test data.

Question 10. How do residual plots help in diagnosing a linear regression model?

Answer: A residual plot shows residuals(the difference between the actual values and the predicted values for the target variable) on the y-axis and the predicted value on the x-axis. It helps visualize how well the model fits the data. it allows us to check for linearity and heteroscedasticity . a residual plot should be centered around zero , residuals should be randomly scattered and there should be no obvious pattern.